FUNDAMENTAL DATA TYPES

BASICS OF C++

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Identifiers

Fundamental Data Types

Compound Data Types

IDENTIFIERS

IDENTIFIERS

A **valid identifier** is a sequence of one or more letters, digits, or underscore (_).

number

age_in_year

result_1

The C++ language is a "**case sensitive** language"

The **Result** variable is **not** the same as the **result** variable

Reserved keywords cannot be use as identifiers

refer to C++ documentation for the complete list

class

auto

catch

final

CODING EXERCISE

```
#include <iostream>
using namespace std;
int main( ) {
    int age = 18;
    cout << "My age is";
    cout << age;
    return 0;
}
```

FUNDAMENTAL DATA TYPES

CHARACTER

The most basic type is **char**, which is a **one-byte** character

Represent a **single** character, such as 'A' or '$'

```
#include <iostream>
using namespace std;
int main( ) {
    char middle = 'A';
    ----------------------
    ----------------------
}
```

NUMERICAL

Store a **whole number** value, such as 7 or 1024

Can either be **signed** or **unsigned**

```
#include <iostream>
using namespace std;
int main( ) {
    int age = 18;
    ----------------------
    ----------------------
}
```

FLOATING POINT

Represent **real values**, such as 3.14 or 0.01

```
#include <iostream>
using namespace std;
int main( ) {
    float weight = 51.8;
    ----------------------
    ----------------------
}
```

BOOLEAN

Can only represent one of two states, **true** or **false**

```
#include <iostream>
using namespace std;
int main( ) {
    bool result = true;
    ----------------------
    ----------------------
}
```

COMPOUND DATA TYPES

STRING CLASS

Store **sequences** of **characters**, such as words or sentences.

```
#include <iostream>
#include <string>
using namespace std;
int main( ) {
    string name = "Gyro";
    string last_name = "Madrona";
    cout << "Hello, My first name is ";
    cout << name << endl;
    cout << "My last name is ";
    cout << last_name;
    ------------------------------------
    ------------------------------------
```

LABORATORY